LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Approaches to Engine Health Monitoring, Various Engine Health Monitoring techniques and Concept of Engine Health Monitoring as part of QA
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Approaches to Engine Health Monitoring, Various Engine Health Monitoring techniques and Concept of Engine Health Monitoring as part of QA
5. Specific Objectives	26.1.1.1 Trainee will be able to explain the engine health monitoring techniques, Purpose and advantages

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Engine Health Monitoring Techniques; Safety: No live engine operation; Instructor supervision: Monitor trainee engagement and Q&A; flow
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engine Health Monitoring?
(b) List two advantages of implementing EHM in aviation maintenance.
(c) How does EHM differ from traditional scheduled maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Engine Health Monitoring • Purpose of EHM in Aviation • Advantages of EHM over Scheduled Maintenance • Approaches to Engine Health Monitoring • Various EHM Techniques Overview	L & Ds	PPT, IB	15 min
(b)	• Detailed Analysis of Vibration Monitoring • Overview of Oil Analysis Techniques • Principles of Acoustic Emission Monitoring • Role of Thermography in EHM • Integration of Multiple Monitoring Techniques	L & Ds	PPT, PDF	20 min
(c)	• Concept of EHM within Quality Assurance • Data Management and Reporting Standards • Predictive Maintenance Strategies • Impact on Operational Readiness • Future Trends in Engine Health Monitoring	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how EHM contributes to Quality Assurance protocols.
(b) Describe one specific technique used for detecting engine anomalies.
(c) How does predictive maintenance reduce downtime compared to scheduled maintenance?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts learned in this lesson will be applied in the upcoming lesson on MCD testing for engine health monitoring.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using MCD test	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min